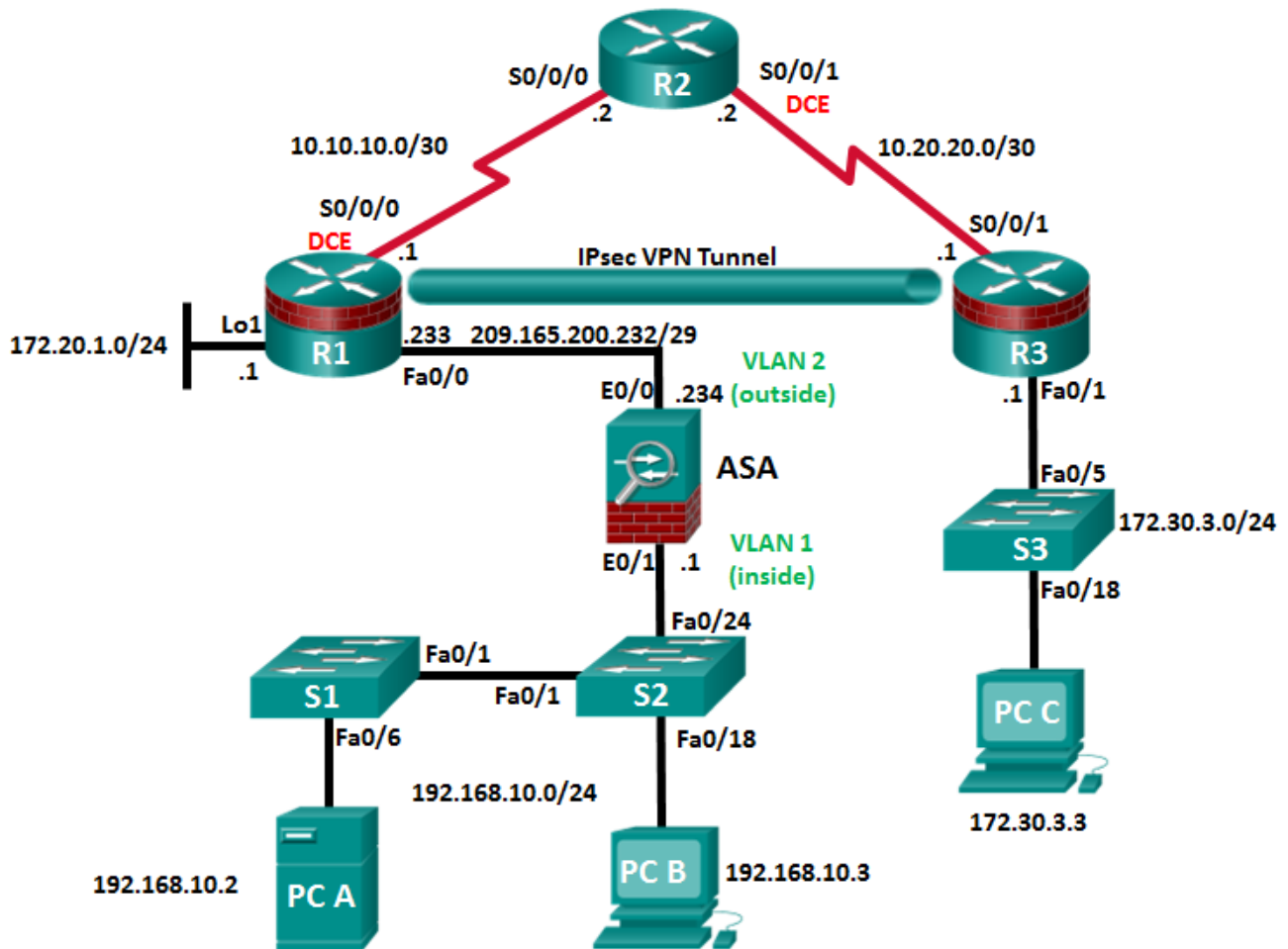


Topology (A EMULER AVEC EV-NG OU GNS3)



IP Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	Fa0/0	209.165.200.233	255.255.255.248	N/A	ASA E0/0
	S0/0/0 (DCE)	10.10.10.1	255.255.255.252	N/A	N/A
	Loopback 1	172.20.1.1	255.255.255.0	N/A	N/A
R2	S0/0/0	10.10.10.2	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.20.20.2	255.255.255.252	N/A	N/A
R3	Fa0/1	172.30.3.1	255.255.255.0	N/A	S3 Fa0/5
	S0/0/1	10.20.20.1	255.255.255.252	N/A	N/A
S1	VLAN 1	192.168.10.11	255.255.255.0	192.168.10.1	N/A
S2	VLAN 1	192.168.10.12	255.255.255.0	192.168.10.1	N/A
S3	VLAN 1	172.30.3.11	255.255.255.0	172.30.3.1	N/A
ASA	VLAN 1 (E0/1)	192.168.10.1	255.255.255.0	N/A	S2 Fa0/24
ASA	VLAN 2 (E0/0)	209.165.200.234	255.255.255.248	N/A	R1 Fa0/0
PC-A	NIC	192.168.10.2	255.255.255.0	192.168.10.1	S1 Fa0/6
PC-B	NIC	192.168.10.3	255.255.255.0	192.168.10.1	S2 Fa0/18
PC-C	NIC	172.30.3.3	255.255.255.0	172.30.3.1	S3 Fa0/18

Objectives

Part 1: Verify Basic Device Settings

Part 2: Configure Secure Router Administrative Access

- Configure encrypted passwords and a login banner.
- Configure EXEC timeout on console and VTY lines.
- Configure login failure rates and VTY login enhancements.
- Configure Secure Shell (SSH) access and disable Telnet.
- Configure local authentication, authorization, and accounting (AAA) user authentication.

Part 3: Configure a Site-to-Site VPN between ISRs

- Configure an IPsec site-to-site VPN between R1 and R3 using the Cisco Configuration Professional

Part 4: Configure a Zone-Based Policy Firewall and Intrusion Prevention System

- Configure a Zone-Based Policy Firewall (ZPF).
- Configure an Intrusion Prevention System (IPS).

Part 5: Secure Network Switches

- Configure passwords and a login banner.
- Configure management VLAN access.
- Secure trunk ports.
- Secure access ports.

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- Protect against Spanning Tree Protocol (STP) attacks.
 - Configure port security and disable unused ports.

Part 6: Configure ASA Basic Settings and Firewall

- Configure basic settings, passwords, date, and time.
- Configure the inside and outside VLAN interfaces.
- Configure Port Address Translation (PAT) for the inside network.
- Configure a Dynamic Host Configuration Protocol (DHCP) server for the inside network.
- Configure administrative access via Telnet and SSH.
- Configure a static default route for the Adaptive Security Appliance (ASA).
- Configure local AAA user authentication.
- Verify address translation and firewall functionality

Part 7: Configure ASA Clientless SSL VPN Remote Access

- Configure a remote access SSL VPN using the Cisco Adaptive Security Device Manager (ASDM).
- Verify SSL VPN access to the portal.

Note: The router commands and CCP output in this SA are from Cisco 1841 routers with Cisco IOS Release 15.1(4)M8 (Advanced IP image). The switch commands and output are from Cisco WS-C2960-24TT-L switches with Cisco IOS Release 15.0(2)SE4 (C2960-LANBASEK9-M image). The ASA is a 5505 with OS version 8.4(2) and ASDM version 7.2(1). Other routers, switches, and Cisco IOS versions may be used. See the Router Interface Summary table at the end of this SA to determine which interface identifiers to use based on the equipment in this SA. Depending on the router or switch model and Cisco IOS version, the commands available and output produced might vary from what is shown in this SA.

Required Resources

- 3 routers (Cisco 1841 with Cisco IOS Release 15.1(4)M8 Advanced IP Services or comparable)
- 3 switches (Cisco 2960 with Cisco IOS Release 15.0(2)SE4 C2960-LANBASEK9-M image or comparable)
- 1 ASA 5505 (OS version 8.4(2) and ASDM version 7.2(1) and Base license or comparable)
- PC-A: Windows Vista, or Windows 7 with CCP, PuTTY SSH client (Web and FTP server optional) (flash drive optional)
- PC-B: Windows Vista, or Windows 7 with PuTTY SSH client and latest Java version. (ASDM loaded on the PC is optional)
- PC-C: Windows Vista, or Windows 7 with CCP, PuTTY SSH client, TFTP server and IPS files. (flash drive optional)
- Serial and Ethernet cables as shown in the topology
- Console cables to configure the routers, switches and ASA via the console

Part 1: Verify Basic Device Settings

Important Note: Prior to starting the exam, verify that the basic test environment is built as described below. Contact your instructor if the following devices and settings have not been pre-configured.

Step 1: Network is cabled as shown in the topology.

Step 2: Basic settings for each router are configured.

- a. Hostname, interface IP addresses, and serial interface DCE clock rate.
- b. DNS lookup disabled.

Step 3: Static default routes on routers R1, R2, and R3 are configured.

- Static default route from R1 to R2 and from R3 to R2.
- Static routes from R2 to the R1 simulated LAN (Loopback 1), the R1 Fa0/0-to-ASA subnet and the R3 LAN.

Step 4: Basic settings for each switch are configured.

- Hostname, VLAN 1 management address, and IP default gateway.
- DNS lookup disabled.

Step 5: Each PC host IP settings are configured.

Static IP address, subnet mask, and default gateway for each PC.

Step 6: Verify connectivity between PC-C and R1 Lo1 and Fa0/0.

Part 2: **Configure Secure Router Administrative Access**

Task 2: Configure Settings for R1 and R2.

Step 1: Configure a minimum password length of 10 characters on R1.

Step 2: Configure the enable secret password on R1.

Use an enable secret password of **ciscoenapa55**.

Step 3: Encrypt plaintext passwords on R1.

Step 4: Configure the console line on R1.

Configure a console password of **ciscoconpa55** and enable login. Set the **exec-timeout** value to log out after 15 minutes of inactivity. Prevent console messages from interrupting the command entry.

Note: The vty lines for R1 are configured for SSH in Task 3.

Step 5: Configure a login warning banner on R1.

Configure a warning to unauthorized users with a message-of-the-day (MOTD) banner that says:
Unauthorized access strictly prohibited and prosecuted to the full extent of the law!

Step 6: Configure the vty lines and enable password on R2.

- Configure the password for vty lines to be **ciscovtypa55** and enable login. Set the **exec-timeout** value so a session is logged out after 15 minutes of inactivity.
- Use an enable secret password of **ciscoenapa55**.

Step 7: Enable HTTP access on R2.

Enable the HTTP server on R2 to simulate an Internet target for later testing.

Task 3: Configure Local Authentication with AAA on R1.

Step 1: Configure the local user database.

Create a local user account of **Admin01** with a secret password of **Admin01pa55** and a privilege level of **15**.

Step 2: Enable AAA services.

Step 3: Implement AAA services using the local database.

Create the default login authentication method list using case-sensitive local authentication as the first option and the enable password as the backup option to use if an error occurs in relation to local authentication.

Task 4: Configure the SSH Server on R1.

Step 1: Configure the domain name: ccnasecurity.com.

Step 2: Configure the incoming vty lines.

Specify that the router vty lines will accept only SSH connections.

Step 3: Generate the RSA encryption key pair.

Configure the RSA keys with **1024** as the number of modulus bits.

Step 4: Configure the SSH version.

Specify that the router accept only SSH version 2 connections.

Step 5: Verify SSH connectivity to R1 from PC-C.

Launch the SSH client on PC-C and test SSH connectivity to R1 and log in in as **Admin01** with the password **Admin01pa55**.

Task 5: Secure against login attacks on R1.

Step 1: Configure enhanced login security on R1.

If a user fails two login attempts within a **30** second time span, then disable logins for **1** minute. Log all failed login attempts.

Part 3: **Configure a Site-to-Site IPsec VPN between ISRs**

In Part 3 of this SA, you use CCP to configure an IPsec VPN tunnel between R1 and R3 that passes through R2.

Task 6: Configure the site-to-site VPN between R1 and R3.

Step 1: Configure the enable secret password and secure HTTP server on R3 prior to starting CCP.

- From the CLI, configure an enable secret password of **ciscoenapa55** for use with CCP on R3.
- Enable the secure HTTP server on R3.
- Add user **Admin01** to the local database with a privilege level of **15** and a password of **Admin01pa55**.
- Configure the local database authentication of HTTP sessions.

Step 2: Access CCP and discover R3.

From PC-C, run CCP and access R3.

Step 3: Use the CCP VPN wizard to configure R3.

Use the **Quick Setup** option to configure the R3 side of the site-to-site VPN.

Step 4: Configure basic VPN connection information settings.

- Specify **R3** interface **S0/0/1** as the interface for the connection and R1 interface **S0/0/0** as the remote peer static IP address.
- Specify the pre-shared VPN key **cisco12345**.
- Encrypt traffic between the R3 LAN and the R1 Loopback 1 simulated LAN.

Step 5: Generate a mirror configuration from R3 and apply it to R1.

- On R3, generate a mirror configuration for application to R1 and save it to the desktop or flash drive. Edit the file as necessary and apply the mirrored commands to R1.
- Apply the crypto map to the R1 VPN interface.

Task 7: Test the Site-to-Site IPsec VPN using CCP

Step 1: On R3 (PC-C), use CCP to test the IPsec VPN tunnel between the two routers.

Step 2: Ping from PC-C to the R1 Lo1 interface at 172.20.1.1 to generate some interesting traffic.

Step 3: Issue the `show crypto isakmp sa` command on R3 to view the security association created.

Step 4: Issue the `show crypto ipsec sa` command on R1 to verify packets are being received from R3 and decrypted by R1.

Part 4: **Configure a Zone-Based Policy Firewall and Intrusion Prevention System**

In Part 4, you configure a ZPF and IPS on R3 using CCP.

Task 8: Configure a ZPF on R3 using CCP.

Step 1: Access CCP and discover R3 (if required).

Step 2: Use the CCP Firewall wizard to configure a Zone-Based Policy Firewall.

- Configure a **Basic** firewall with Fa0/1 interface as the **Inside** interface and S0/0/1 as the **Outside** interface.
- Use the **Low Security** setting, and complete the Firewall wizard.

Step 3: Verify Firewall functionality.

- From PC-C, ping external R2. The pings should be successful.
- From external R2, ping PC-C. The pings should NOT be successful.
- From **PC-C**, Telnet to R2 at IP address **10.20.20.2** and use password **ciscovtpa55**.
- With the Telnet session open from PC-C to R2, issue the **show policy-map type inspect zone-pair session** command on R3. Continue pressing **Enter** until you see an Established Sessions section toward the end.

Part 5: **Secure Network Switches**

Task 9: Configure Passwords and a Login Banner on S1.

Step 1: Configure the enable secret password of ciscoenapa55.

Step 2: Encrypt plaintext passwords.

Step 3: Configure the console and VTY lines.

- a. Configure a console password of **ciscoconpa55** and set the **exec-timeout** value to log out after 5 minutes of inactivity. Prevent console messages from interrupting command entry.
- b. Configure a vty line password of **ciscovtypa55** and set the **exec-timeout** to log out after 5 minutes of inactivity.

Step 4: Configure a login warning banner.

Configure a warning to unauthorized users with a MOTD banner that says: **Unauthorized access strictly prohibited and prosecuted to the full extent of the law!**

Step 5: Disable HTTP access.

HTTP access to the switch is enabled by default. To prevent HTTP access, disable the HTTP server and HTTP secure server.

Task 10: Secure Trunk and Access Ports on S1 and S2.

Step 1: Configure trunk ports on S1 and S2.

Step 2: Change the native VLAN to 99 for the trunk ports on S1 and S2.

Step 3: Prevent the use of DTP on S1 and S2 trunk ports.

Step 4: Verify the trunking configuration on S1 and S2.

Step 5: Enable storm control for broadcasts on S1 and S2 trunk ports.

Step 6: Disable trunking on S1 access ports that are in use.

Step 7: Enable PortFast on S1 access ports that are in use.

Step 8: Enable BPDU guard on S1 access ports that are in use.

Task 11: Configure Port Security and Disable Unused Ports.

Step 1: Configure basic port security for the S1 access port.

Use the default port security options (set maximum MAC addresses to 1 and violation action to shutdown). Allow the secure MAC address that is dynamically learned on a port be added to the switch running configuration.

Step 2: Disable unused ports on S1.

As a further security measure, disable any ports not being used.

Step 3: Save the running-config to the startup-config for each switch.

Part 6: Configure ASA Basic Settings and Firewall

Task 12: Prepare the ASA for ASDM access.

Step 1: Clear the previous ASA configuration settings.

Step 2: Bypass Setup Mode and configure the VLAN interfaces using the CLI.

- a. The VLAN 1 logical interface will be used by PC-B to access ASDM on ASA physical interface E0/1. Configure interface VLAN 1 and name it **inside**. Specify IP address **192.168.10.1** and subnet mask **255.255.255.0**. Verify that the security level is set to **100**.
- b. Pre-configure interface VLAN 2 and name it **outside**, and add physical interface E0/0 to **VLAN 2**. You will assign the IP address using ASDM. Verify that the security level is set to **0**.
- c. Test Connectivity to the ASA by pinging from PC-B to ASA interface VLAN 1 IP address **192.168.10.1**. The pings should be successful.

Step 3: Configure and verify access to the ASA from the inside network.

- a. Configure the ASA to accept HTTPS connections and to allow access to ASDM from any host on the inside network 192.168.10.0/24.
- b. Open a browser on PC-B and test the HTTPS access to the ASA ASDM GUI.

Task 13: Configure basic ASA settings using the ASDM Startup Wizard.

Step 1: Access the Configuration menu and launch the Startup wizard.

Step 2: Configure the hostname, domain name, and enable password.

Configure the ASA host name **CCNAS-ASA** and domain name of **ccnasecurity.com**. Change the enable mode password to **ciscoenapa55**.

Step 3: Configure the outside VLAN interface.

Enter an outside IP address of **209.165.200.234** and mask **255.255.255.248**.

Step 4: Configure DHCP, address translation, and administrative access.

- a. Enable the DHCP server on the Inside Interface and specify a starting IP address of **192.168.10.5** and ending IP address of **192.168.10.30**. Enter the DNS server 1 address of **10.3.3.3** and domain name **ccnasecurity.com**.
- b. Configure the ASA to use PAT using the IP address of the outside interface.
- c. Add Telnet access to the ASA for the inside network **192.168.10.0** with a subnet mask of **255.255.255.0**. Add SSH access to the ASA from host **172.30.3.3** on the outside network.

Step 5: Test Telnet access to the ASA.

From a command prompt or GUI Telnet client on PC-B, telnet to the ASA inside interface at IP address 192.168.10.1.

Task 14: Configuring ASA Settings from the ASDM Configuration Menu.

Step 1: Set the ASA Date and Time.

Set the time zone, current date and time, and apply the commands to the ASA.

Step 2: Configure a static default route for the ASA.

Add a static route for the outside interface and specify **Any** for the network object and a Gateway IP of **209.165.200.233**. Use **Ping** from the ASDM **Tools** menu to test connectivity to the IP address of R1 S0/0/0 (10.10.10.1). The ping should succeed.

Step 3: Test access to an external website from PC-B.

Open a browser on PC-B and enter the IP address of the R2 S0/0/0 interface (**10.10.10.2**) to simulate access to an external website. The R2 HTTP server was previously enabled so you should be prompted with a user authentication login dialog box from the R2 GUI device manger. Exit the browser.

Step 4: Configure AAA for SSH client access.

- Create a new user named **admin** with a password of **cisco123**. Allow this user **Full access** (ASDM, SSH, Telnet, and console) and set the privilege level to **15**.
- Require authentication for HTTP/ASDM, SSH, and Telnet connections and specify the **LOCAL** server group for each connection type.
- From PC-C, open an SSH client, such as PuTTY, and attempt to access the ASA outside interface at **209.165.200.234**. You should be able to establish the connection.

Part 7: **Configure ASA Clientless SSL VPN Remote Access**

Step 5: Start the VPN wizard.

Step 6: Configure the SSL VPN user interface.

Configure **VPN-Con-Prof** as the Connection Profile Name, and specify **outside** as the interface to which outside users connect.

Step 7: Configure AAA user authentication.

Use the local user database to authenticate remote access users and create a new user named **VPNuser** with a password of **remote**.

Step 8: Configure the VPN group policy.

Create a new group policy named **VPN-Grp-Pol**.

Step 9: Configure the bookmark list

- Add a bookmark list and name it **WebServer-XX** (where XX is your initials).
- Add a new Bookmark with **Web Mail** as the Bookmark Title. Specify the server destination IP address of PC-B **192.168.10.3** (simulating a web server).

Step 10: Verify VPN access from the remote host.

Open the browser on PC-C and enter the login URL for the SSL VPN into the address field (**https://209.165.200.234**). The Logon window should appear. Enter the previously configured user name **VPNuser** and password **remote** and click **Logon** to continue. The Web Portal window should display.
